assignment - circular2 and intro torque

This is a preview of the published version of the quiz

Started: Apr 29 at 8pm

Quiz Instructions

there is no tutorial but hints. **BONUS AHEAD!!!**

Remember to use a textbook. Try hard before getting the answers.

Question 1 10 pts

A new roller coaster contains a loop-the-loop in which the car and rider are completely upside down. If the radius of the loop is 13.2m with what minimum speed must the car traverse the loop so that the rider does not fall out while upside down at the top? Assume the rider is not strapped to the car. Hide hints first.

hint; see my slides 98/99. At the top, weight is down and normal is down (from track to roller coester). If the cart barely touches the track, N = 0 so you have: weight (mg)= centripetal force (mV^2/R). Solve for speed. you see that the mass cancell out.

- 10.1 m/s
- 12.5 m/s
- 11.4 m/s
- 14.9 m/s

Question 2 10 pts

A 2.0-kg ball is moving with a constant speed of 5.0 m/s in a horizontal circle whose diameter is 1.0 m. What is the magnitude of the net force on the ball?

hint: radius is half the diameter. The net force is the centripetal force.

- 0 N

50 N

40 N

20 N

100 N

Question 3 10 pts

An object moves with a constant speed of 30 m/s on a circular track of radius 150 m. What is the acceleration of the object?

hint: its the centripetal acceleration

0.17 m/s^2

6.0 m/s^2

0 m/s^2

5.0 m/s^2

Question 4 10 pts

A point on a wheel of radius 40 cm that is rotating at a constant 5.0 revolutions per second is located 0.20 m from the axis of rotation. What is the acceleration of that point due to the spin of the wheel? (try without hint)

Hint: The speed of the rim is not the same as the speed of the point. (a point on the rim covers more distance in the same time) But they have the same period (time it takes to go around). 5 revolutions per second = 5 x 2pi (0.2) m per second. That's the speed. Then apply the equation a = V^2/0.2. So don't use the 40cm/ Its Tmi

0.00 m/s^2

200 m/s^2

48 m/s^2

○ $0.050\ m/s^2$

○ $1.4\ m/s^2$

Question 5 5 pts

What is the SI unit of torque?

○ Joule (J)

○ Newton (N)

○ Watt (W)

○ Newton-meter (N.m)

Question 6 5 pts

If you apply a force of 20 Newtons at a distance of 5 meters from the pivot point, what is the torque?

○ 5N.m

○ 20N

○ 100 N.m

○ 4N.m

Question 7 5 pts

In a seesaw, where should a heavier person sit to balance the seesaw with a lighter person?

○ The torque doesn't depend on weight distribution

○ It doesn't matter; both can sit anywhere

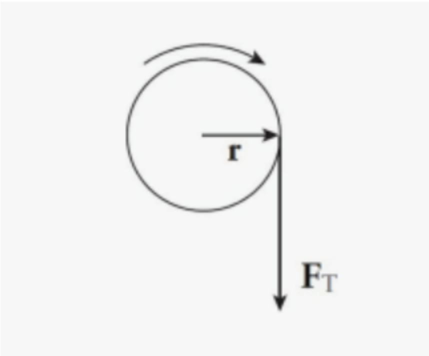
○ Closer to the pivot point

○ Farther from the pivot point

Question 8 10 pts

A student pulls down with a force of 40 N on a rope that winds around a pulley of radius 5 cm. see picture

What's the torque of this force? (hint: **CW is negative torque** and CCW is positive torque)

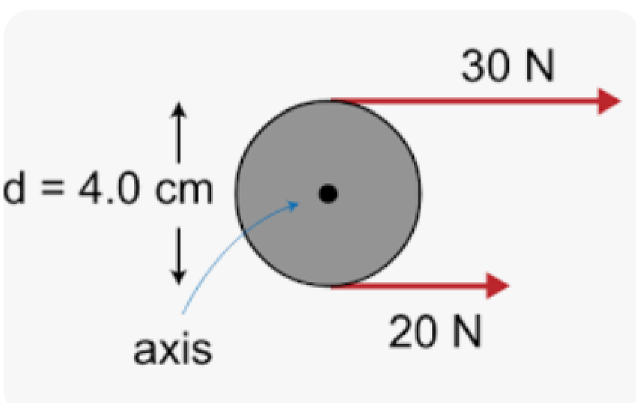


- +2 N.m
- + 200N.m
- - 2N.m
- -200 N.m

Question 9 10 pts

The cylinder below is free to rotate around its center. What is the net torque on the cylinder?

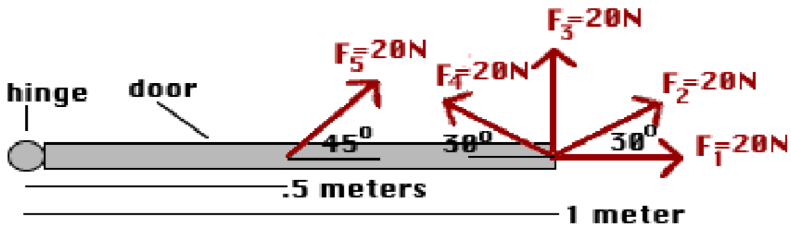
hint: The radius is 2cm. convert to meters. a CW torque is negative and a CCW is positive. Find the torque applied by each force. Is it negative or positive? (one is + and one is -) Then you add. Convert to m



- -0.2 N.m
- -0.6 N.m
- 1 N.m

0.4N.m

Question 10 5 pts



What is the torque applied by F5 ?

hint: CCW is positive. Find the perpendicular component of F5. and multiply by the distance between the point where the force is applied and the axis.

10 N.m

7 N.m

-7 N.m

14N.m

Question 11 10 pts

Building on the previous question. What is the force applied by F1?

hint:is F2 efficient

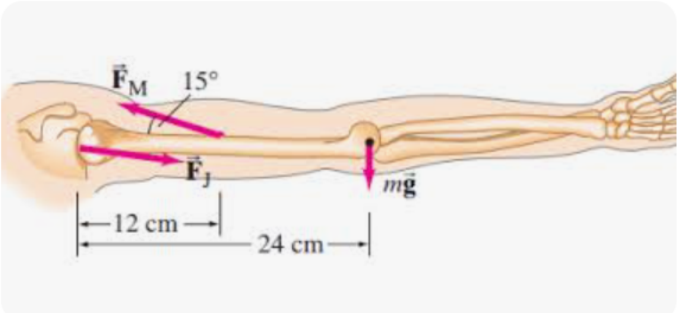
-20N.m

10N.m

20 N.m

0 N.m

Question 12 10 pts



the magnitude of the force M (deltoid muscle) is 1840N. The distance between the shoulder joint and the point where M is applied is 15cm (the distance between the axis and the tail of the vector M).

What is the torque applied by Fm about the shoulder (axis of rotation) ?

hint: you need to find the perpendicular componet of FM. perpendicular to the arm. convert cm to m

- 476N.m
- 220 N.m
- 57N.m
- 5715N.m

Question 13 5 pts

One revolution is the same as

- pi/2 rad
- 1 rad
- 57 rad
- 2 *pi rad

Question 14 1 pts

The angular speed of the second hand of a watch is

(hint: think !)

- (π/60) rad/s
- (π/30) rad/s

(2π) rad/s

(60) rad/s

Question 15 5 pts

A circle has a diameter of 4m. The distance around is about

25.1m

3.14m

12.6m

6.3m

Not saved Submit Quiz